

DOUALA, Cameroon

WMO: 649100

Lat: 4.006N

Lon: 9.719E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.7	73.0	70.1	110.7	77.6	71.3	115.5	76.2	13.3	81.7	12.0	81.6	2.1	80	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.1	93.1	80.7	91.7	80.5	91.3	80.4	83.8	88.2	82.6	87.7	81.8	87.2	6.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.7	170.7	85.8	82.0	166.6	85.3	80.6	159.1	84.3	48.0	89.6	46.6	86.3	45.6	89.2	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.5	9.3	70.0	95.3	1.2	1.0	69.1	96.0	68.4	96.5	67.8	97.1	66.9	97.8		
(5)			69.1	85.1	1.4	2.5	68.0	86.9	67.2	88.3	66.4	89.7	65.3	91.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	83.4	84.0	83.4	82.6	82.2	80.4	78.7	78.1	79.3	80.3	81.6	83.0
	DBStd	2.63	1.42	1.82	2.01	2.22	2.04	1.80	1.72	1.56	1.81	1.81	2.01	1.39
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11464	1036	951	1035	978	999	913	889	873	880	939	949	1022
	CDD65	5989	571	531	570	528	534	463	424	408	430	474	499	557
	CDH74	55804	6044	5953	6272	5477	5418	4035	2862	2401	3096	3850	4667	5729
	CDH80	18482	2361	2430	2547	2137	1977	1033	429	262	608	1051	1527	2120
Wind	WSAvg	2.9	2.6	3.1	3.1	3.0	2.9	2.8	2.9	3.2	3.0	2.8	2.6	2.3
Precipitation	PrecAvg	150.2	1.3	2.2	6.7	9.5	10.8	16.6	25.4	28.5	23.9	15.8	5.4	1.4
	PrecMax	192.3	6.3	8.7	13.7	21.1	16.2	31.1	47.6	48.8	40.4	24.9	13.9	4.9
	PrecMin	102.2	0.0	0.0	1.9	4.9	6.0	6.9	10.5	14.8	10.4	7.7	0.3	0.0
	PrecStd	20.9	1.4	2.0	3.0	3.1	2.8	5.4	9.1	8.3	8.0	4.0	3.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	93.6	93.6	93.5	93.0	89.9	87.5	86.1	88.2	90.0	91.6	92.1
		MCWB	79.4	80.7	81.3	81.5	81.2	79.9	78.8	78.5	79.0	79.8	80.9	79.6
	2%	DB	91.7	92.8	92.4	91.8	91.3	88.2	85.7	84.3	86.4	88.2	89.9	91.3
		MCWB	79.6	80.5	81.0	81.2	80.8	79.7	78.2	77.7	78.6	79.5	80.3	79.8
	5%	DB	91.0	91.4	91.3	91.0	89.8	87.0	84.1	82.6	85.6	87.4	89.0	89.9
		MCWB	79.6	80.3	80.7	80.9	80.3	79.2	77.5	77.2	78.3	79.3	80.1	79.6
	10%	DB	89.5	89.9	89.7	89.4	88.2	85.6	82.5	81.0	83.9	85.8	87.6	89.2
		MCWB	79.4	80.1	80.3	80.4	80.0	78.6	77.1	76.9	77.6	78.7	79.9	79.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.7	84.5	84.6	86.1	84.0	81.0	79.7	80.5	80.2	82.4	83.2	83.3
		MCDB	87.8	88.7	88.5	89.0	89.0	87.0	84.5	82.4	85.2	85.3	86.8	87.7
	2%	WB	82.1	82.8	82.8	83.3	82.0	80.1	78.9	78.8	79.3	80.6	81.7	81.4
		MCDB	87.3	87.8	88.1	88.2	88.1	85.8	83.5	82.2	84.3	86.0	86.3	87.3
	5%	WB	81.0	81.7	81.8	81.8	81.0	79.6	78.2	78.0	78.7	79.7	80.7	80.6
		MCDB	86.9	87.1	87.5	87.4	87.1	84.9	82.4	81.2	83.5	84.9	86.5	87.1
	10%	WB	80.2	80.9	81.0	80.9	80.3	79.0	77.6	77.3	78.1	79.0	80.0	80.0
		MCDB	86.1	86.6	86.8	86.7	86.0	83.8	81.4	80.3	82.3	83.9	85.4	85.9
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.1	14.2	14.1	13.5	10.8	8.3	7.5	9.8	12.0	12.9	13.8
		MCDBR	15.5	14.8	15.5	15.9	15.1	13.0	10.5	9.4	12.0	14.2	14.4	14.7
		MCWBR	5.2	5.0	6.1	6.7	6.4	5.4	4.3	4.2	5.0	6.3	6.0	5.1
	5% WB	MCDBR	14.3	14.3	14.5	14.7	14.2	11.7	9.2	8.5	10.6	13.0	13.6	13.8
		MCWBR	5.5	5.8	6.5	7.5	6.6	5.3	4.4	4.4	4.9	6.5	6.2	5.3
Clear-Sky Solar Irradiance	taub	0.803	0.832	0.805	0.570	0.503	0.524	0.522	0.507	0.450	0.467	0.587	0.739	
	taud	1.525	1.495	1.543	2.016	2.213	2.141	2.146	2.193	2.369	2.335	1.950	1.623	
	Ebn at Noon	189	189	195	242	252	243	245	254	274	271	237	200	
	Edh at Noon	93	98	93	57	45	48	48	47	40	42	60	83	
All-Sky Solar Radiation	RadAvg	1601	1521	1469	1486	1436	1195	1046	1011	1170	1302	1402	1538	
	RadStd	82	92	87	49	65	107	88	68	71	54	58	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page